

Height = 20'
Mean Height = 25'

$$q_z = (0.00256)(.99)(1)(1)(1)(15 \text{ mph})^2$$

$$K_{15} = 28.75 \text{ Avg Pressure for 25ft}$$

$$K_{20} = 30.47 = 20' \text{ Height pressure}$$

$$K_{30} = 33.179 = 30' \text{ Height pressure}$$

$$K = P = q \cdot G \cdot C_p - q_h(G C_p)$$

Windward wall Case 2a

$$q_{20} = 30.47, C_p = 0.8$$

$$(30.47)(.85)(.8) = 20.72 \text{ PSF}$$

$$(28.75)(.85)(.8) = 19.57$$

Leeward Wall

$$(31.82)(.85)(-.46) = -12.44$$

Side walls

(interpolated)

$$q_h = 31.82 \text{ PSF } C_p = -3.11$$

$$(31.82)(.85)(-.7) = -18.93$$

Windward Roof

$$(31.82)(.85)(-.31) = -8.41$$

Leeward Roof

$$(31.82)(.85)(-.6) = -16.23$$

$$L/B = \frac{48}{40} = 1.2$$

$$H/L = \frac{25}{48} = .52$$

Length =

Width =

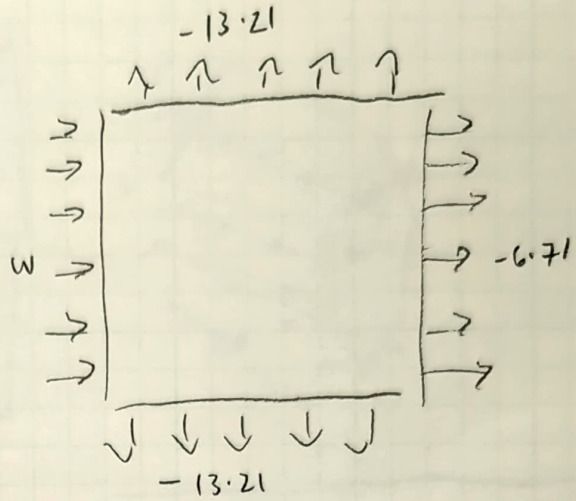
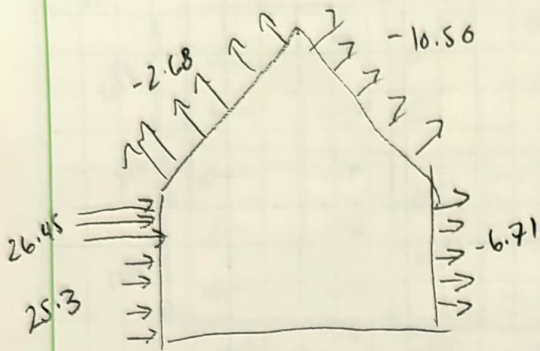
Height = 25'

$$(q_h)(G C_p) = (31.23)(.18) = 5.73 \text{ PSF}$$

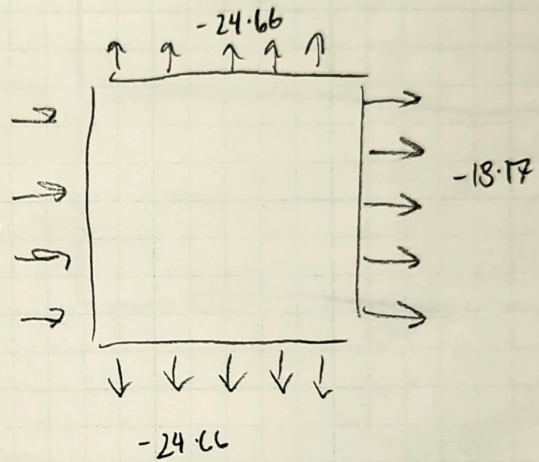
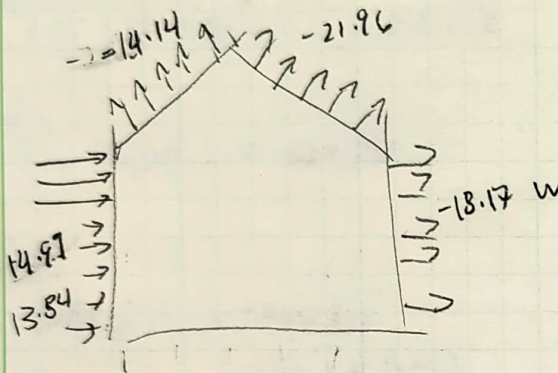
Case 1 +0.18	Case 2 -0.18
26.45	14.99
25.3	13.84
-6.71	-18.17
-13.21	-24.66
-2.68	-14.14
10.5	-21.96

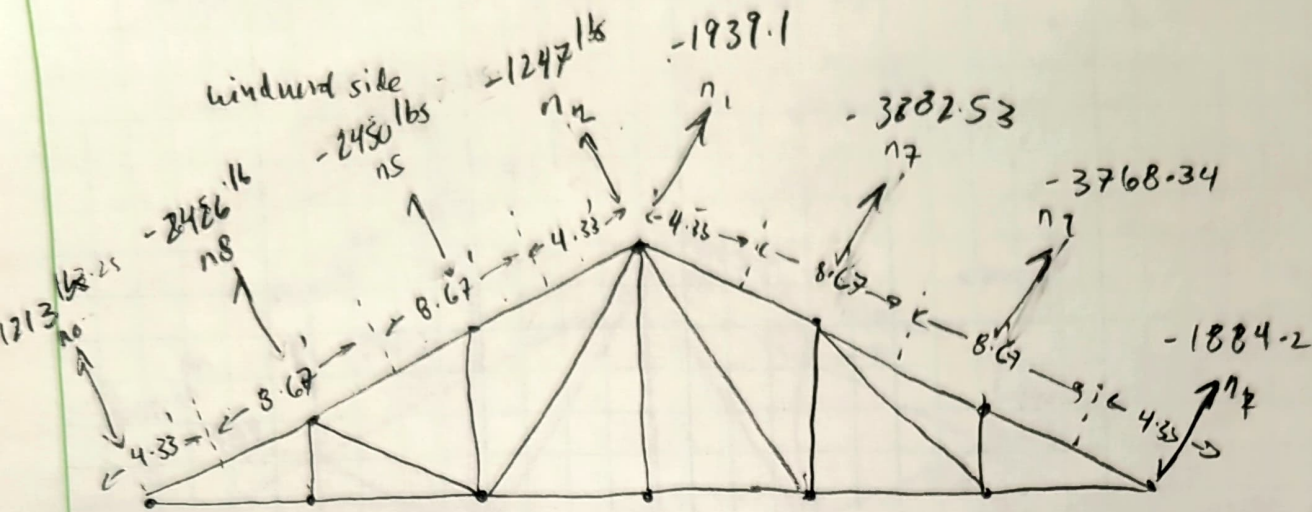
Reaction Forces

Case 1



Case #2





windward loads (Roof)

$$(4.29)(20)(14.14) = -1213$$

$$(8.58)(20)(14.14) = -2426$$

$$(8.84)(20)(14.14) = -2456$$

$$(4.415)(20)(14.14) = -1247$$

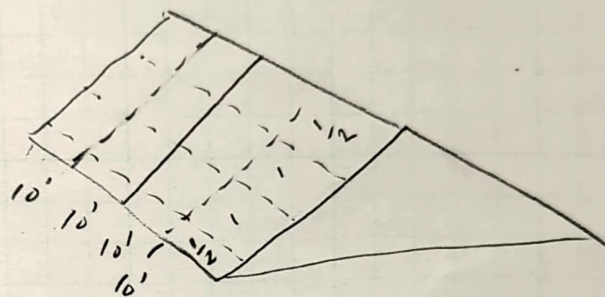
leeward loads (Roof)

$$(4.29)(20)(-21.96) = -1884.2$$

$$(8.58)(20)(-21.96) = -3768.34$$

$$(8.84)(20)(-21.96) = -3882.53$$

$$(4.415)(20)(-21.96) = -1939.1$$



$$\begin{matrix} \textcircled{S6} & \textcircled{S7} \\ -5.8 & -1245.785 \end{matrix}$$

For dead loads see previous spreadsheet

For snow loads see previous spreadsheet

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$$\begin{matrix} f_{.32} = f_x \text{ or } a \\ f_{.92} = f_y \text{ or } \beta \end{matrix} \quad S7, S7, G1$$

$$\begin{matrix} x & y & x & x & x & y & x & y & x & y & x & y & x & y \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{matrix}$$

$$(1.2DL) + 1LL + 1WL + .5SL$$

$$-0.538$$

Pins at N_0, N_2, N_3

25, 26, 27

Forces and loads on Nucleus, Question 70:

